CE 206: Engineering Computation Sessional

Numerical Differentiation and Integration

`diff` and `gradient` command

`diff(x)`

Returns the difference between adjacent elements in x. Typically used for unequally spaced intervals

`fx = gradient(f, h)`

Determines the derivative of the data in *f* at each of the points. *h* is the spacing between points; if omitted h=1.

Uses forward difference for the first point, backward difference for the last point, and centered difference for the interior points.

`diff` and `gradient` Example

```
function y=fx(x)
y=0.2+25*x-200*x.^2+675*x.^3-900*x.^4+400*x.^5;

x =[0 0.12 0.22 0.32 0.36 0.4 0.44 0.54 0.64 0.7 0.8];
y=fx(x);
```

Derivative at unequally spaced intervals:

```
diff(x) ./diff(y)
```

Derivative at equally spaced intervals from 0 to 0.8

```
x =0:0.1:0.8;
y=fx(x);
gradient(y,0.1)
```

Example: Newton's law of cooling

The rate of cooling is proportional to the differences in temperature of the body and the surrounding medium. This can be expressed as:

$$\frac{dT}{dt} = -k(T - T_a)$$

Where, T = temp. of the body, T_a = temp. of the surrounding medium, k = a proportionality constant (per min)

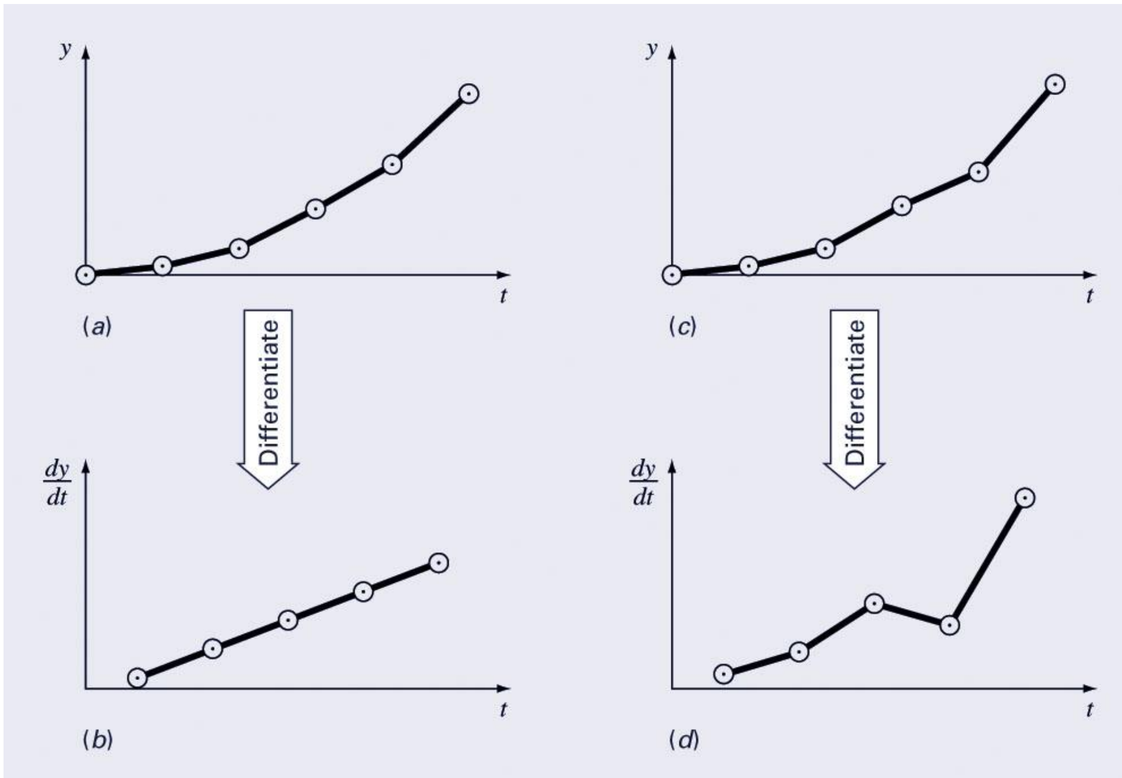
If a metal ball heated to 80°C is dropped into water having T_a = 20°C, the temp of the ball changes as in

Time (min)	0	5	10	15	20	25
T (°C)	80	44.5	30	19.1	21.7	20.7

(a) Determine *dT/dt* using numerical differentiation
(b) Using linear regression, determine *k*.

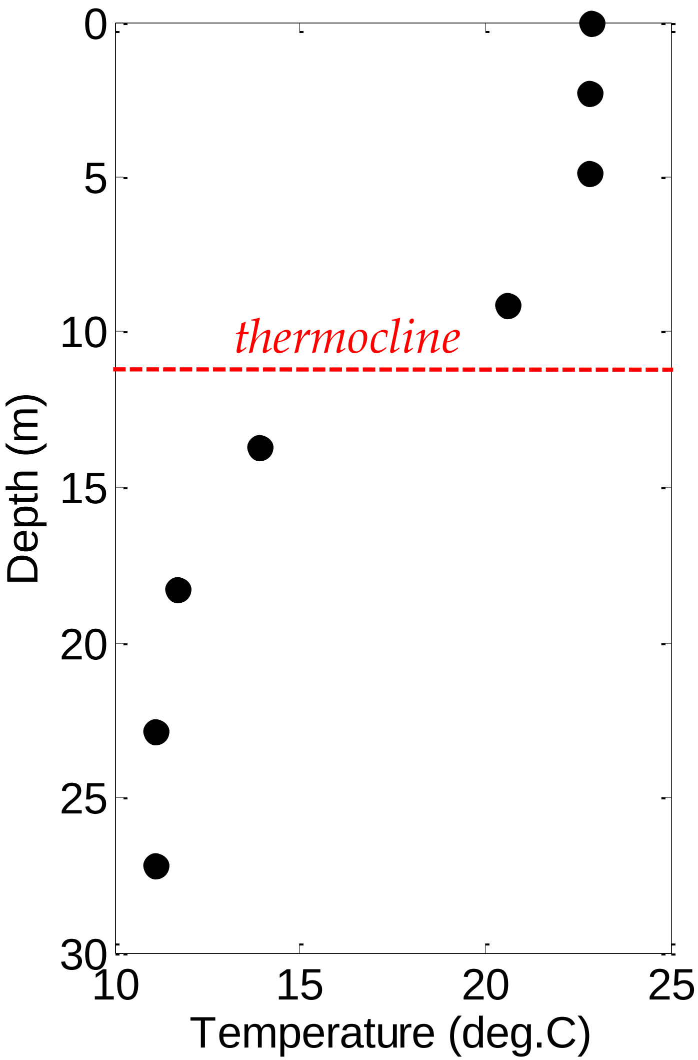
Derivatives and Errors

A shortcoming of numerical differentiation is that it tends to amplify errors in data



How to minimize this effect: fit a smooth, differentiable function to the data and take the derivative of the function.

Assignment prob: Thermocline in a lake



General knowledge:

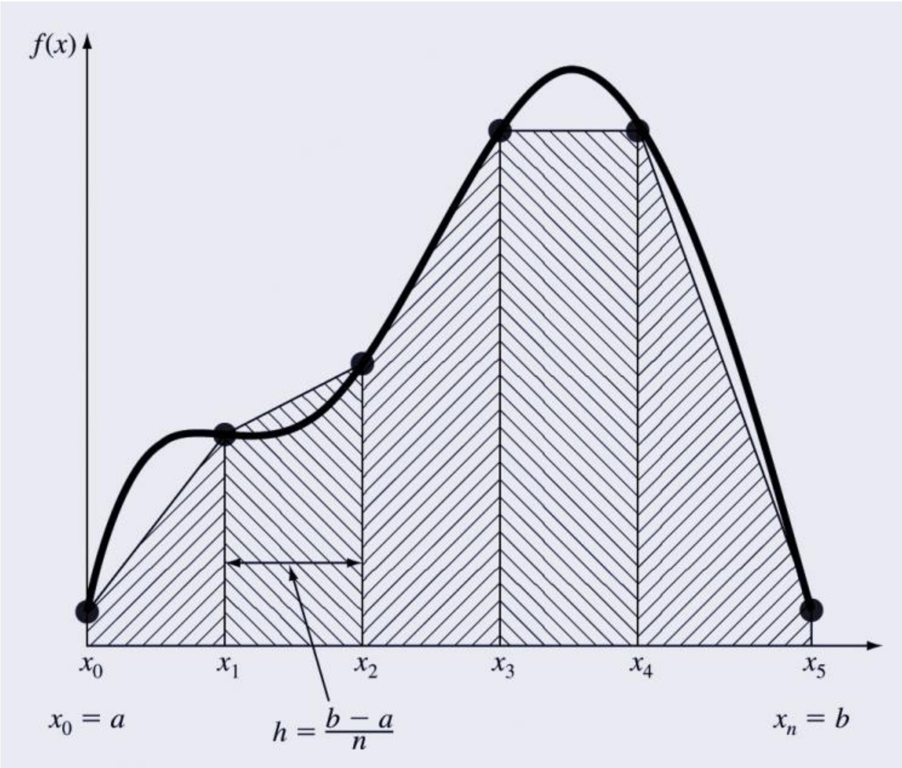
What is a thermocline? Why is it important in a lake environment?

Question:

How to locate the thermocline?

The thermocline is located at the inflexion point of the temperature depth curve (where $d^2T/dx^2 = 0$). It is also the point where dT/dx is maximum.

Trapezoidal Rule



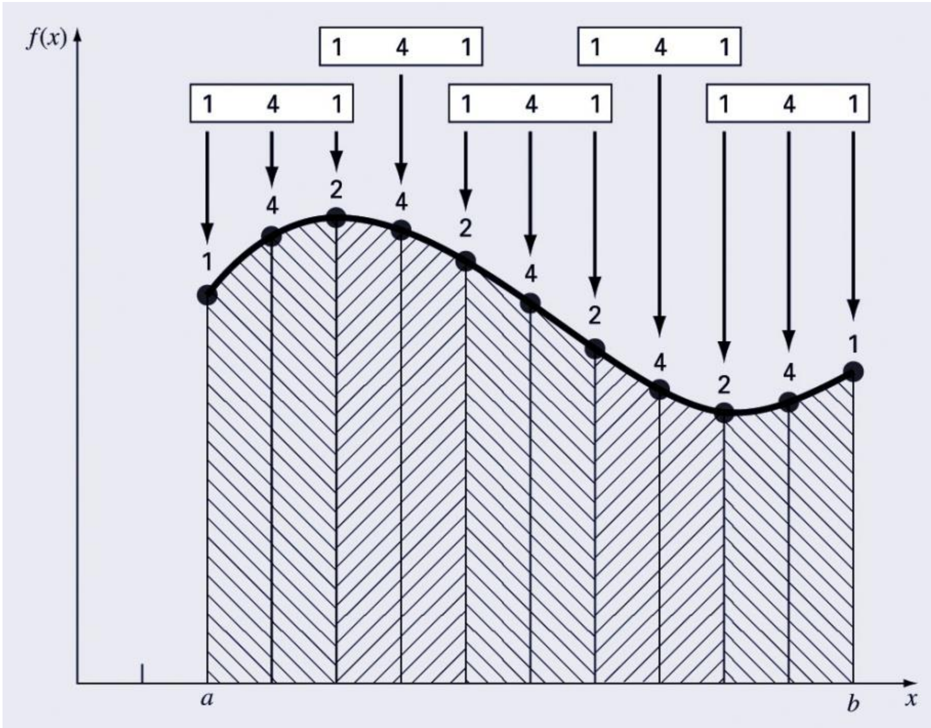
Assuming n+1 data points are evenly spaced, there will be n intervals over which to integrate.

$$I = \frac{h}{2}\left[f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n) \right]$$

For unequal segments

$$I = (x_1 - x_0)\frac{f(x_0) + f(x_1)}{2} + (x_2 - x_1)\frac{f(x_1) + f(x_2)}{2} + \cdots + (x_n - x_{n-1})\frac{f(x_{n-1}) + f(x_n)}{2}$$

Simpson's 1/3 Rule



Can be used for an odd number of points

$$I = \frac{h}{3}\left[f(x_0) + 4f(x_1) + f(x_2)\right] + \frac{h}{3}\left[f(x_2) + 4f(x_3) + f(x_4)\right] + \cdots + \frac{h}{3}\left[f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\right]$$

$$I = \frac{h}{3}\left[f(x_0) + 4\sum_{\substack{i=1 \\ i,\,\text{odd}}}^{n-1} f(x_i) + 2\sum_{\substack{j=2 \\ j,\,\text{even}}}^{n-2} f(x_i) + f(x_n) \right]$$

MATLAB `trapz` function

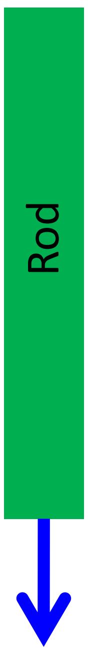
Built-in functions to evaluate integrals based on trapezoidal rule

```
z = trapz(y)
z = trapz(x, y)
```

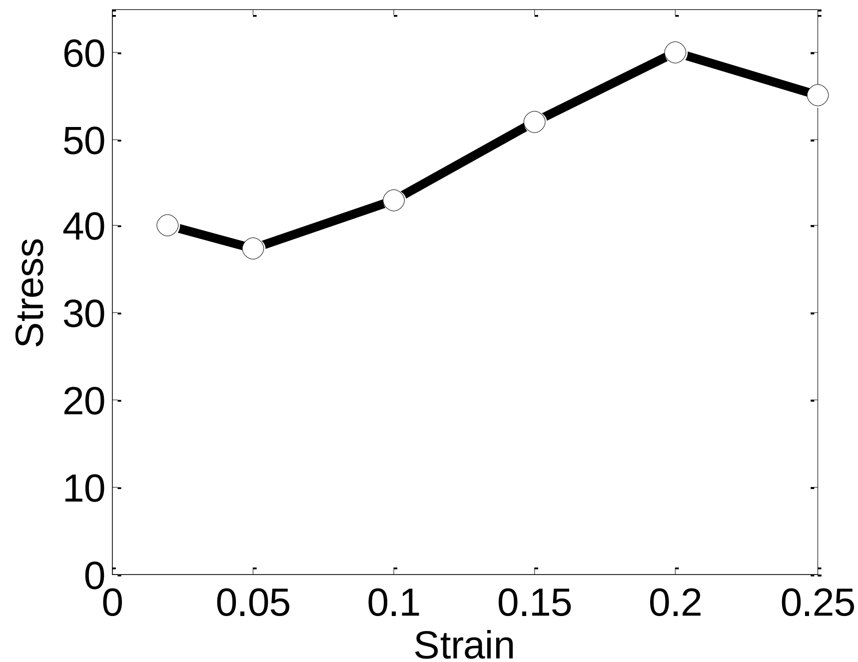
produces the integral of y with respect to x. If x is omitted, the program assumes h=1.

Ex: Modulus of toughness from σ-ε curve

A rod subjected to an axial load yielded the following data upto rupture



ε	σ (kip/in^2)
0.02	40.0
0.05	37.5
0.10	43.0
0.15	52.0
0.20	60.0
0.25	55.0

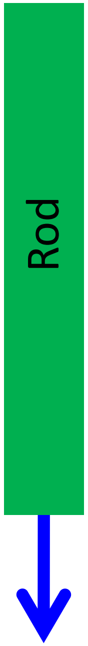


The area under the stress-strain diagram upto rupture is called the <u>modulus of rupture</u>

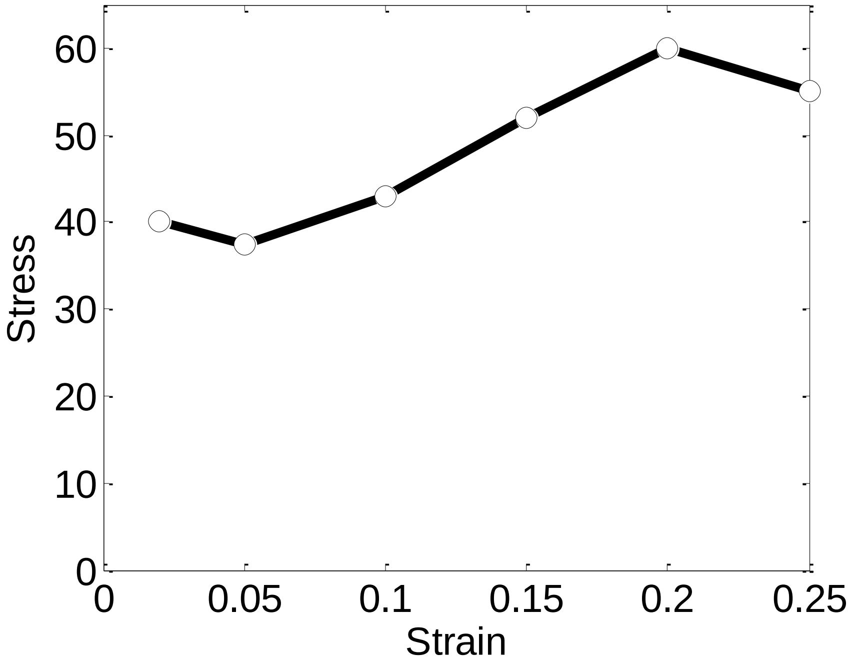
```
>> trapz(e,s)

ans =
   11.2250
```

Class exercise



ε	σ (kip/in^2)
0.02	40.0
0.05	37.5
0.10	43.0
0.15	52.0
0.20	60.0
0.25	55.0



Redo the same problem in the following way:

(a) Fit a polynomial with the stress-strain data. Show the plot.

(b) Use trapezoidal rule using 100 equal segments, the ordinates of these points evaluated from the fitted polynomial

Exercise: Work done by a force

Work done = force × distance

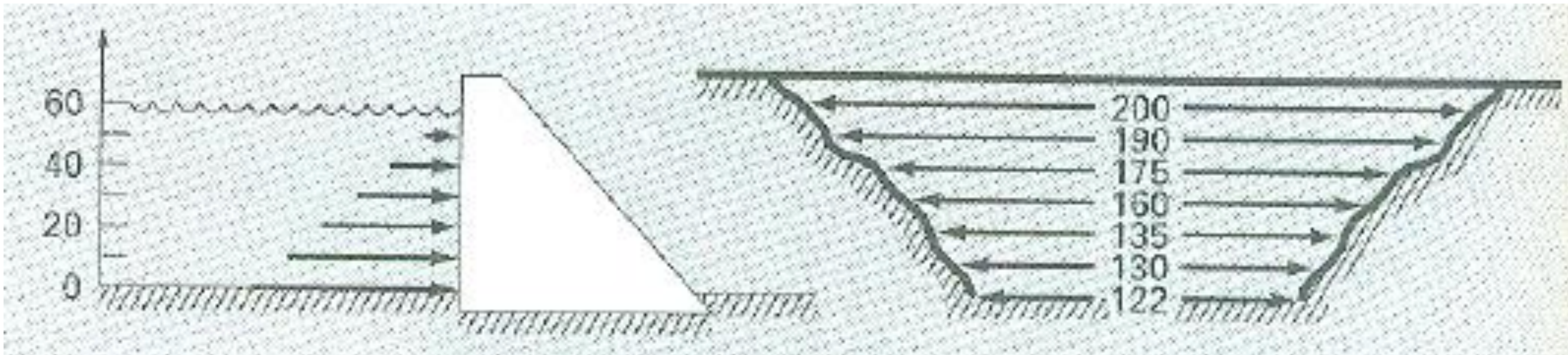
Calculate the work done from the following discrete measurements of F(x) and θ(x)

$$W = \int F(x)dx$$

X (ft)	F(x), lb	Θ (rad)
0	0	0.50
5	9	1.40
10	13	0.75
15	14	0.90
20	10.5	1.30
25	12	1.48
30	5	1.50

Exercise: Solving problems in hydrostatics

Calculate the force exerted on the dam and the line of action



ρ = 1000 kg/m^3

g = 9.81 m/sec^2

D = 60 m

w = width

$$P = \int_0^D \rho g w(z)(D - z)dz$$

$$d = \frac{\int_0^D \rho g z w(z)(D - z)dz}{\int_0^D \rho g w(z)(D - z)dz}$$

<u>Ans</u>: $P = 2.54 \times 10^9$ N

$d = 21.97$ m

Exercise: Integrals of functions

The total force exerted on the mast of a sailboat can be expressed as the integral of the following continuous function:

$$F = \int_0^{30} \underbrace{200\left(\frac{z}{5+z}\right)e^{-2z/30}}_{\text{Nonlinear integral, difficult to evaluate analytically}} dz$$

Evaluate the integral using trapezoidal rule using step sizes 15, 10, 6, 3, 1, 0.5, 0.25, 0.1 ft

[Ans: 1001.7 lb, 1222.3 lb, 1372.3 lb 1450.8 lb, 1477.1 lb 1479.7 lb, 1480.3 lb 1480.5 lb 1480.6 lb]